Information

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Team F:

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ILR01

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Individual Progress

Since our group was formed, I would discuss ideas with the group over the two meetings we had. While we would each start our ideas, I would think of both mechanisms and sensors to add for the robot, leading to the three sensors I would choose for the lab, the potentiometer, joystick, and 9 DOF IMU. Both the potentiometer and joystick are currently planned to go together for the hand held controller to direct movement for the hexapod, while the IMU is being planned to be used as a “self checker” for movement.

Challenges

For each sensor, I would have a challenge.

For the potentiometer, while the mechanism itself worked seamlessly first try, I was originally debating using a potentiometer or a rotary encoder as the sensor to control power. With the rotary encoder, while it detected each revolve, I was having it detect spinning the other way and later would end up scrapping that idea to go fully with the potentiometer. If an issue of access to analog pins arrises, I will reconsider going back.

For the joystick, the main issue I would encounter was with “zeroing” the joystick. At rest, the joystick would not account the x and y axis of the joystick to be at (0,0), but instead some value between +- 5 on both x and y axis. Talking to others, supposedly there is a way to zero the joystick, but in lack of time, there would just be a threshold to pass to cause movement.

For the IMU, there was two main issues. For this lab, the plan was to only use some of the data to tell generally direction. Collecting the other data off the IMU resulted in funky values, which I was unable to tell whether it was from my code or the IMU being messed up or glitched. We were using an already used IMU from previous years found in the closet in a random box not placed and packaged safely. In the end, the test code for telling direction would pass on the IMU and work correctly.

One more issue I would run into from the sensor lab was with experimenting with the Arduino Megas that roboclub has and I have access to as a Senior Member. I wanted to implement a Mega instead of an Uno due to having access to more pins, specifically analog pins. The first board would work perfectly with everything up to the IMU, when I would realize that the I2C pin was shot. For the second board, I would have issues until realizing that I was unable to push code at all to the other Mega. Debugging the board issues would take up most of my time in the lab until I would call quits on both and switch back to an Arduino Uno. In the end, there was barely enough Analog pins for all 3 sensors.

Teamwork

Due to the nature of the snow, most teammates were unable to make it physically to help or go over the lab. However, while working on the lab, we would meet virtually to go over sensors and

over future steps for the project. Also, Thomas would work on refining my code after I finished physically setting up the lab, turning values like numbers to actual data with blocking. For example, instead of having a value like 376 from the potentiometer, it would list “Speed Level: 3”.

Plans

Currently, my plan is to meet with the team regarding different sketches seen below in Figure 1: alongside creating more prototypes, both in drawing and in CAD, of different things to consider mechanical and electronically for the Hexapod. Furthermore, after meeting with each member, I would realize that I had the most experience with hexapods in general from research, and will be finding references and resources for the members to help explain to them the different areas of hexapods.

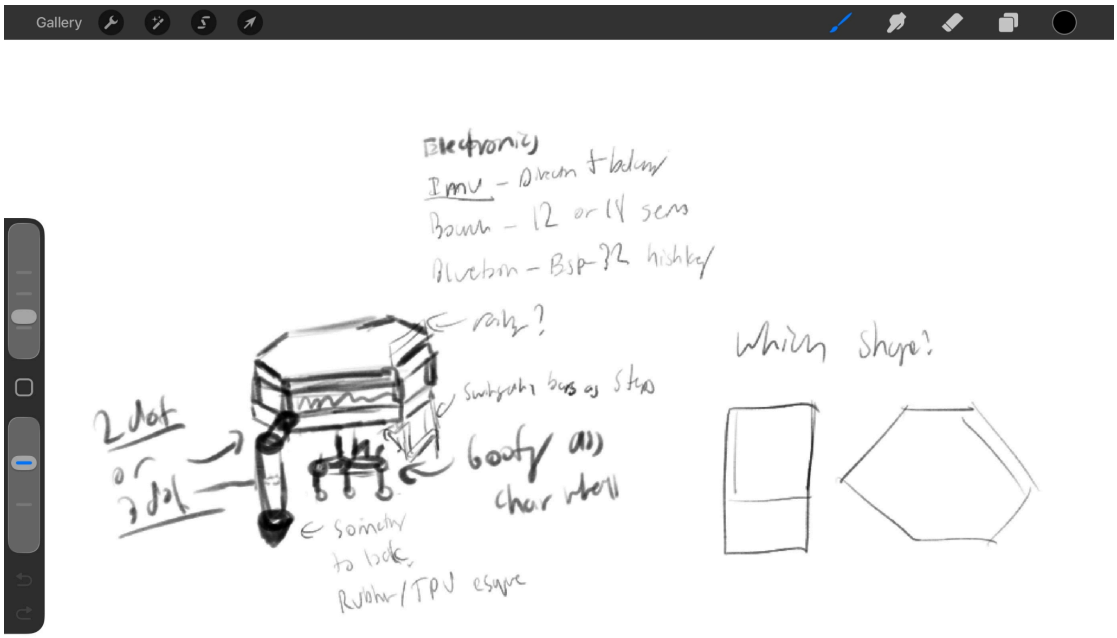

Figure 1: Sketches of different parts and prototyping for hexapod